First work package: backend foundation

Two weeks · by Kunal Balwani · a proposal to adapt at the kickoff, not a fixed spec

> **Goal:** by the end of week 2, your iOS and Android developers are building against a real, deployed staging API, and every permission rule is proven by a test that runs on every pull request.

Day by day

Day	What gets built	What you see or decide
Week 1 · foundation and permissions		
Day 1	Kickoff call. Repository, Supabase staging and production projects (EU region), CI pipeline running on every pull request.	30-min kickoff; a written decisions note the same day
Day 2	First migrations: people, horses, the roles on each horse, stables and stable members.	Data model diagram to review
Day 3	One permission function used by every policy; row-level security on every table.	**Permission matrix**: who may do what, for you to approve
Day 4	Permission test suite: each rule tested as owner, trainer, caretaker, stable staff, manager and a stranger.	Test report in plain language
Day 5	API contract v1 (read views, one function per write) and the contract document generated from the database.	Contract handed to the mobile developers for comments
Week 2 · the stable day-to-day, offline and handover		
Day 6	Training logs and stable tasks, with ids created on the phone so a retried save never duplicates.	Mobile developers can create and list both
Day 7	Sensor readings: idempotent batch upload, monthly partitions, upload watermark per device.	Readings demo with a batch sent twice
Day 8	Delta sync with deletions (tombstones), so a phone that was offline catches up correctly.	Offline scenario tests added to CI
Day 9	Deployment pipeline: staging on every merge, production with a manual approval. Seed data for the app teams.	Staging URL and test accounts
Day 10	Handover session with the mobile developers; request examples for every endpoint; plan for package 2.	Written summary and package 2 proposal

Day numbers are working days. If something takes longer, I tell you the same day, with the reason and the new estimate.

Done means

✓ Every table has row-level security switched on, and no write happens without the permission check.

✓ All permission and sync tests pass in CI, and a pull request that breaks one cannot be merged.

✓ The contract document matches the deployed database; CI fails if they drift apart.

✓ The iOS and Android developers can sign in to staging with seed accounts and call every endpoint in the contract.

✓ Every decision taken along the way is written down where the team can find it.

Decisions I need from you

☐ **Stable hierarchy:** does a stable manager see every horse boarded there, while staff see only their own tasks?

☐ **Product passport:** readable by anyone who scans the QR code, or only by signed-in owners?

☐ **Hosting region:** confirm the EU region for staging and production.

☐ **Reviews:** who reviews pull requests on your side, and who signs off the permission matrix?

☐ **Channel:** where you want daily updates (Slack, Teams, GitHub or here).

How you will hear from me

Every working day

- A short written update: done, next, blocked.
- Small pull requests, each with a plain description.
- Questions asked when something is unclear, not guessed.

Every week

- A Friday summary for you and the app teams.
- A demo of what works on staging.
- The plan for the next week, with anything that changed.

Not in this package

Planned for package 2, so the foundation is solid first: QR product passport and ownership transfer, integrations (outbox and webhooks), photo and media storage with signed links, push notifications, and the admin side.

Why start this way

Permissions and the API contract are the two things that are expensive to change later and that your mobile developers depend on first. Getting them right, tested and in their hands within two weeks lets the rest of the work run in parallel, and gives you a concrete way to judge my work on your real project before committing further.